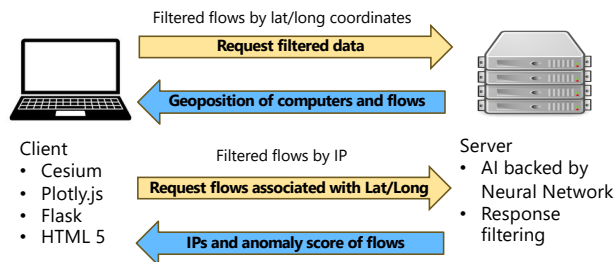


AI-Driven Network Anomaly Detection & Visualization

Abstract

Visualization of anomalous network flows across the entire world. Highly unusual and anomalous data flows are detected using AI and are highlighted in red. Clicking on circles and lines provides additional flow information. This project has many real-world applications like stopping hacking, DDOS attacks, and theft of intellectual property.

Methods



Client: A Flask app was used to send the filter settings to the server every time it updated. The server would respond with a list of nodes and edges to draw. It also sends back a list of data points to draw on the chart. The client then promptly draws them, where nodes and edges with more connections are bigger, and ones with higher anomaly scores are more red. When a node or edge is clicked, the client sends the node and filter settings to the server. The server sends back a list of IPs and their corresponding anomaly scores.

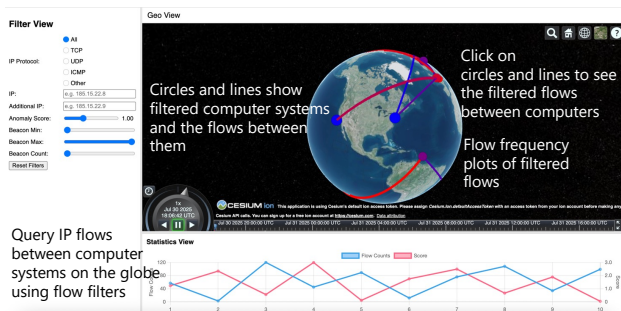
A variety of software is used in this project. Cesium renders the 3D globe. Plotly creates the chart. HTML makes the webpage. Flask allows server-client communication.

Server: Retrieves a dataset and filters based on client requests. It assigns an anomaly score based on a Neural Network Machine Learning model, which was previously trained.

It takes the input flow (a vector of features describing the network/activity flow), compresses it into a compact representation via the encoder, then expands it back via the decoder to produce an output vector that is aimed to match the original. That output is the “reconstruction” of the flow.

If the autoencoder has learned the typical patterns, normal flows will be reconstructed closely (small differences), whereas unusual flows will normally produce poor reconstructions (large differences). The difference between the original and reconstructed flow is the reconstruction error.

Results & Impact



The visualization greatly helps to report and filter network anomalies, allowing users the flexibility to apply filters and zero in on areas of interest. Primary accomplishments include:

- Filtering for IP Protocol, IP, additional IP, and anomaly score
- Created Flask App and interactive UI
- Reduced server response time by 86%
- Fixed many bugs (e.g., filters)

Ongoing & Future Research

- Additional reporting capabilities, including dashboards and user-configurable reports
- Enable encryption to enhance security
- Utilize client and server-side caching for faster response times
- Add more filters like pre-burst time and burst time
- Leverage on-premise cloud to process large datasets on GPUs
- Improve current model performance in terms of accuracy and training time
- Evaluate other models to enhance evaluation scores
- Generate synthetic data of high quality to be used for model training



NAVAL
POSTGRADUATE
SCHOOL

SEAP Student Researcher: Brady Liang
Advisor: Bruce Allen